# FORM 2

*(See Section 10 and Rule 13)*

# COMPLETE SPECIFICATION

## TITLE OF INVENTION:

## A MULTI-DOMAIN CYBER-PHYSICAL INTRUSION DETECTION AND RESPONSE SYSTEM FOR ELECTRIC POWER DISTRIBUTION NETWORKS USING PHYSICS-INFORMED NEURAL NETWORKS AND MULTI-LAYER ANOMALY FUSION

| Field | Details |
|---|---|
| **Applicant Name** | Sarin Prem Jaiswal |
| **Nationality** | Indian National |
| **Address** | [Complete Address with PIN Code], India |
| **Inventor(s)** | [FULL LEGAL NAME OF INVENTOR], [Address] |
| **Application Type** | Complete Specification (following Provisional Application No. ____/____/2026) |
| **Date of Filing Provisional** | [Date of Provisional Filing] |
| **Priority Date** | [Date of Provisional Filing] |
| **Field of Technology** | Industrial Cybersecurity; Cyber-Physical Systems; Power Systems Engineering; Artificial Intelligence |
| **Patent Agent** | [Name and Registration Number of Patent Agent, if applicable] |
| **Corresponding Address** | [Address for correspondence] |

## 1. FIELD OF INVENTION

The present invention relates to the field of cybersecurity for industrial control systems and cyber-physical systems, and more particularly to a multi-domain intrusion detection and automated response system for electric power distribution networks. The invention employs Physics-Informed Neural Networks (PINNs), graph neural networks, industrial network protocol analysis, and a multi-layer anomaly fusion engine to detect, classify, and respond to cyber-physical attacks including False Data Injection (FDI), electricity theft, unauthorised SCADA command injection, device impersonation, and coordinated load manipulation — attacks that are undetectable by conventional network-layer intrusion detection systems.

## 2. BACKGROUND OF INVENTION

Electric power distribution networks are a class of cyber-physical systems (CPS) in which digital control commands and sensor data streams have direct physical consequences — circuit breaker operations, voltage regulation, load dispatch, and protection relay coordination. The deployment of SCADA (Supervisory Control and Data Acquisition) systems, smart meters, intelligent electronic devices (IEDs), and advanced metering infrastructure (AMI) has created extensive digital attack surfaces within previously isolated physical infrastructure.

The prior art in operational technology (OT) cybersecurity consists predominantly of network-layer intrusion detection systems that monitor industrial protocol traffic — including Modbus TCP, DNP3, IEC-61850 GOOSE/MMS, and OPC-UA — for protocol anomalies, unauthorised commands, and behavioural deviations. These systems, including commercially deployed platforms such as those described in US Patent Nos. 10,609,068 (network anomaly detection for industrial control systems) and 11,223,637 (ICS threat intelligence correlation), operate exclusively in the cyber domain and have no capability to validate whether reported sensor measurements are physically consistent with the governing laws of the power network.

This fundamental limitation was formally established by Liu, Ning, and Reiter (2011), who proved mathematically that a False Data Injection attack vector satisfying the condition $c = Ha$ — where $H$ is the measurement Jacobian of the linearised power flow model — is completely undetectable by any chi-squared residual test applied to network-layer data. This proof demonstrates that network-layer IDS is provably insufficient for detecting an entire class of power grid attacks. An attacker with knowledge of the grid topology can craft false sensor readings that look entirely legitimate on the network while causing the state estimator to compute a fundamentally incorrect operating point, potentially leading to incorrect dispatch decisions, protection relay miscoordination, and cascading failures.

Electricity theft via smart meter tampering — meter bypassing, current transformer shunting, firmware manipulation — constitutes a second major class of attack that is also undetectable by network-layer monitoring. In India, Non-Technical Losses (NTL) attributable to electricity theft are estimated at over twenty thousand crore rupees annually, affecting the financial sustainability of distribution utilities.

### 2.1 Limitations of Prior Art

Existing intrusion detection systems for industrial control systems suffer from one or more of the following limitations:

(a) Confinement to the network/cyber domain: existing IDS platforms monitor protocol traffic and device behaviour but perform no validation against physical laws governing the power network.

(b) Inability to detect FDI attacks: attacks crafted to satisfy network-layer plausibility criteria are definitionally undetectable by prior art network-layer IDS systems.

(c) No electricity theft detection: smart meter tamper detection requires comparison of aggregate feeder power injection against the sum of customer meter readings — a physical energy balance check unavailable to network-layer systems.

(d) No multi-domain fusion: prior art systems that combine network and process data do so through signature matching or simple threshold rules, without a physics-grounded fusion model.

(e) No automated physical response: existing systems generate alerts but do not implement automated, physics-informed mitigation actions.

(f) No uncertainty quantification: prior art does not provide confidence intervals on anomaly scores, leading to elevated false positive rates under normal load transients.

There exists, therefore, a need in the art for a system and method that validates observed grid sensor measurements against the physical laws of power flow, integrates this physics-layer validation with network-layer anomaly detection, and fuses signals from multiple physical domains into a unified, explainable, automated intrusion detection and response capability.

## 3. OBJECTS OF THE INVENTION

The primary objects of the present invention are:

(i) To provide a system and method for detecting False Data Injection attacks on electric power distribution networks by computing physics residuals between observed sensor readings and values predicted by Physics-Informed Neural Networks trained on power flow governing equations.

(ii) To provide a system and method for real-time electricity theft detection through automated computation of feeder-level energy balance Non-Technical Loss ratios that cannot be manipulated by tampering with individual smart meter firmware or hardware.

(iii) To provide a system and method for SCADA network intrusion detection covering industrial protocols including Modbus TCP, DNP3, IEC-61850 GOOSE/MMS, and OPC-UA, incorporating per-device behavioural profiling, device timing fingerprinting, and graph-based lateral movement detection.

(iv) To provide a multi-domain anomaly fusion engine that combines physics-domain anomaly scores from the Physics-Informed Neural Network, network-domain anomaly scores from the SCADA IDS engine, and optionally thermodynamic side-channel anomaly scores, into a single confidence-weighted threat score with corroboration bonus.

(v) To provide a system and method for anomaly detection that is robust against sophisticated FDI attacks designed to bypass linearised chi-squared detection, by

enforcing full nonlinear AC power flow equation constraints through Physics-Informed Neural Network residuals.

(vi) To provide explainability of anomaly scores through SHAP (SHapley Additive exPlanations) value decomposition, identifying the dominant contributing feature for each generated alert.

(vii) To provide automated incident classification with mapping to the MITRE ATT&CK for ICS framework, and automated generation of structured incident reports conforming to CERT-In reporting requirements.

(viii) To provide a system deployable in both cloud-connected and edge-computing (offline-capable) configurations, enabling operation in communication-constrained industrial environments.

## 4. SUMMARY OF INVENTION

According to one aspect of the present invention, there is provided a cyber-physical intrusion detection and response system for electric power distribution networks, comprising: a data ingestion module configured to parse industrial network protocol traffic into physical state readings; a physics constraint engine configured to compute expected sensor values by solving AC power flow equations for the network topology; a Physics-Informed Neural Network configured to predict physics-consistent expected sensor states; an anomaly detection module configured to compute a False Data Injection residual and apply threshold detection; a threat identification module configured to classify detected anomalies; and a multi-domain fusion and response engine configured to fuse anomaly signals from multiple detection layers into a unified threat score and execute automated response actions.

According to another aspect of the present invention, there is provided a method for detecting False Data Injection attacks on a power distribution network, comprising: receiving observed sensor readings; computing a physics-consistent expected state using a Physics-Informed Neural Network trained with a composite loss function including a physics residual term; computing a False Data Injection residual as the absolute difference between observed and predicted values; applying a statistical test to the residual; and generating an alert when the residual exceeds a Neyman-Pearson calibrated threshold.

According to a further aspect of the present invention, there is provided a method for real-time electricity theft detection in a smart metered distribution network, comprising: receiving feeder-level power injection measurements; receiving individual smart meter consumption readings; computing a Non-Technical Loss ratio as the difference between feeder injection and the sum of meter readings plus estimated technical losses, normalised by feeder injection; and generating an alert when the Non-Technical Loss ratio exceeds a calibrated threshold.

## 5. BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which form a part of this specification, illustrate embodiments of the invention, and together with the description serve to explain the principles of the invention.

FIG. 1 is a system architecture diagram illustrating the four principal engines of the invention: Dvvalin (SCADA Network IDS Engine), AEthelred (Physics-Informed Anomaly Detection Engine), Brynhildr (Thermodynamic Side-Channel Engine), and Valkyrie (Multi-Domain Fusion and Response Engine).

FIG. 2 is a data flow diagram illustrating the six-module detection pipeline of the AEthelred engine: (1) Data Ingestion, (2) Physics Constraints, (3) PINN Prediction, (4) Anomaly Detection, (5) Threat Identification, and (6) Response and Visualisation.

FIG. 3 is a schematic diagram illustrating the Physics-Informed Neural Network architecture and composite training loss function.

FIG. 4 is a block diagram of the Valkyrie multi-domain fusion engine showing confidence-weighted score combination with corroboration bonus.

FIG. 5 is a graph illustrating physics residual values under normal operation versus under False Data Injection attack, showing the residual amplification ratio.

FIG. 6 is a block diagram illustrating the three deployment configurations: cloud-connected, edge-deployed, and hybrid.

# 6. DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description sets forth specific embodiments of the invention. It will be apparent to those skilled in the art that modifications and variations are possible. All such modifications that fall within the scope of the appended claims and their equivalents are intended to be covered.

## 6.1 System Overview

Referring to FIG. 1, the present invention in a preferred embodiment comprises four cooperating detection engines operating on a target power distribution network, each monitoring a distinct physical domain:

(a) A first engine, referred to herein as the SCADA Network Intrusion Detection Engine (10), monitors the cyber domain — industrial network traffic.

(b) A second engine, referred to herein as the Physics-Informed Anomaly Detection Engine (20), monitors the electrical physics domain — power flow consistency of sensor readings.

(c) A third engine, referred to herein as the Thermodynamic Side-Channel Security Engine (30), monitors the thermodynamic domain — hardware thermal and electromagnetic emission signatures.

(d) A fourth engine, referred to herein as the Multi-Domain Fusion and Response Engine (40), fuses signals from engines (10), (20), and (30) into a unified threat score and drives response actions.

The core inventive principle is that an attacker cannot simultaneously: (i) make network traffic appear legitimate to engine (10); (ii) inject sensor values that satisfy the full nonlinear AC power flow equations enforced by engine (20); and (iii) maintain thermodynamically consistent hardware signatures monitored by engine (30) — without leaving detectable physical inconsistencies across at least one monitored domain. This multi-domain detection approach provides a qualitatively stronger security guarantee than any single-domain approach.

## 6.2 AEthelred — Physics-Informed Anomaly Detection Engine (20)

The Physics-Informed Anomaly Detection Engine (20) implements a six-module detection pipeline as illustrated in FIG. 2.

### 6.2.1 Module 1 — Data Ingestion and Protocol Semantic Mapping

Module 1 (21) receives raw industrial network packets and extracts physical state variables — voltage magnitude (V), voltage angle (theta), active power (P), reactive power (Q), and system frequency (f) — from protocol payloads. In a preferred embodiment, Module 1 uses the Zeek network analysis framework with custom industrial protocol scripts. Supported protocols include Modbus TCP (port 502), DNP3 (port 20000), IEC-61850 GOOSE (EtherType 0x88B8), IEC-61850 MMS (port 102), and OPC-UA (port 4840). The output is a time-indexed stream of physical state readings, each associated with a bus identifier in the network topology.

### 6.2.2 Module 2 — Physics Constraints Ingestion and Anchoring

Module 2 (22) computes the expected physical state of the network by solving the full AC power flow equations for the network topology. In a preferred embodiment, Module 2 employs PyPSA (Python for Power System Analysis) to compute, for each bus i in the network:

*Active power: P_i = SUM_j [ V_i * V_j * (G_ij * cos(theta_i - theta_j) + B_ij * sin(theta_i - theta_j)) ]*

*Reactive power: Q_i = SUM_j [ V_i * V_j * (G_ij * sin(theta_i - theta_j) - B_ij * cos(theta_i - theta_j)) ]*

where V_i and V_j are voltage magnitudes at buses i and j; theta_i and theta_j are voltage angles; G_ij and B_ij are the real and imaginary components of the admittance matrix element Y_ij. These computed values constitute the function h(x_expected), which represents the physics-consistent expected sensor readings used in the FDI residual computation of Module 4.

Module 2 additionally computes the admittance matrix Y_bus, bus voltage limits, feeder thermal limits, and the Non-Technical Loss (NTL) ratio for electricity theft detection:

*NTL_ratio = (P_feeder_in - SUM_k(P_meter_k) - P_estimated_loss) / P_feeder_in*

where P_feeder_in is the measured power entering the feeder from the substation, SUM_k(P_meter_k) is the sum of individual customer meter readings, and P_estimated_loss is the technically estimated resistive line loss. An NTL_ratio exceeding a configurable threshold (default 0.05, representing 5%) indicates either electricity theft or meter tampering. This computation is independent of individual meter data and cannot be defeated by tampering with individual smart meter firmware.

### 6.2.3 Module 3 — PINN Prediction Model

Module 3 (23) implements a Physics-Informed Neural Network (PINN) that predicts the expected physical state of the network as a function of bus identifier and time. The PINN is trained on data collected during normal operation of the target network, with a composite training loss function that simultaneously enforces data fidelity and adherence to the governing physical laws:

*L_total = lambda_d * L_data + lambda_p * L_physics + lambda_b * L_boundary*

*L_data = (1/N_d) * SUM_i [ |u_hat(x_i, t_i) - u_i|^2 ]*

*L_physics = (1/N_r) * SUM_j [ |N[u_hat](x_j, t_j) - f(x_j)|^2 ]*

*L_boundary= (1/N_b) * SUM_k [ |B[u_hat](x_k, t_k) - g(x_k)|^2 ]*

where u_hat(x,t) is the PINN network output; u_i are observed sensor measurements; N[u_hat] is the physics operator applied to the PINN output representing the power flow residual; f(x,t) is the source term; B[u_hat] represents boundary conditions; g(x,t) are boundary values; and lambda_d, lambda_p, lambda_b are weighting coefficients. The physics operator N embeds Kirchhoff's Current Law (KCL) and the AC power flow equations as differentiable constraints.

The PINN architecture in a preferred embodiment comprises a feed-forward neural network with five hidden layers, sixty-four neurons per layer, and hyperbolic tangent activation functions, taking as input the normalised bus identifier and normalised timestamp, and producing as output the predicted normalised voltage magnitude, voltage angle, active power, and reactive power at the specified bus and time.

In an alternative embodiment, Module 3 implements a Graph Attention Network (GAT) that represents the power network as a graph with buses as nodes and transmission

lines as edges, enabling the physics residual computation to exploit topological neighbourhood relationships between buses.

In a further embodiment, Module 3 implements a Bayesian Physics-Informed Neural Network using Monte Carlo Dropout, wherein N stochastic forward passes are performed at inference time to produce a mean prediction and a per-output standard deviation. Detection is triggered only when the observed measurement deviates by more than a configurable number of standard deviations from the predicted mean, substantially reducing false positive rates under normal load transients.

### 6.2.4 Module 4 — Real-Time Anomaly Detection

Module 4 (24) computes the False Data Injection (FDI) residual by comparing the observed sensor reading z_observed from Module 1 against the physics-consistent expected value h(x_expected) derived as the weighted combination of the PINN prediction from Module 3 and the physics oracle from Module 2:

*r(z) = | z_observed - h(x_expected) |*

*h(x_expected) = alpha * u_hat_PINN(x, t) + (1 - alpha) * h_PyPSA(x)*

where alpha is a configurable weighting parameter (default 0.5). This dual-oracle approach makes the reference harder for an attacker to deceive, as it requires simultaneous evasion of both the learned PINN model and the analytically computed physics oracle.

A regression model f is applied to the residual to compute a scaled anomaly score X:

*X = f(r(z))  [where f is a Random Cut Forest or equivalent]*

*V = X^2 > tau  [anomaly flag, tau calibrated by Neyman-Pearson on validation set]*

Additionally, Module 4 applies a chi-squared statistical test for classical FDI detection:

*||r(z)||^2 > chi^2_alpha(m)  [chi-squared with m measurements, significance alpha]*

The PINN-based residual detects sophisticated FDI attacks that satisfy the linearised chi-squared test (the c = Ha bypass condition) because the PINN enforces the full nonlinear AC power flow equations rather than the linearised Jacobian H employed in classical state estimation. This is a key technical advantage of the present invention over prior art.

Module 4 further implements multi-timescale detection: a fast detection window of three consecutive readings (approximately six seconds at default polling interval) for detecting transient FDI attacks; and a slow detection window of 150 readings (approximately five minutes) for detecting gradual electricity theft and slow-ramp load manipulation attacks.

Module 4 additionally computes the voltage stability L-index at each load bus:

*L_j = |1 + SUM_i(F_ji * V_i / V_j)|*

*where F = -[Y_LL]^(-1) * [Y_LG]  (Thevenin equivalent matrix)*

Values of L approaching unity indicate voltage collapse conditions. This provides operational grid health monitoring beyond the primary security function.

### 6.2.5 Module 5 — Threat Identification and Classification

Module 5 (25) classifies detected anomalies into threat categories using a trained classification model operating on feature vectors derived from the residuals of Module 4. Feature vectors are pre-processed using Signal-to-Noise Ratio (SNR) feature filtering, wherein features are ranked by their between-class to within-class variance ratio (F-statistic), and low-SNR features are excluded from the classification input. Outlier training instances are additionally filtered using Interquartile Range (IQR) bounds to improve effective training SNR.

The primary threat categories detectable by Module 5 include: False Data Injection; Electricity Theft; Load Manipulation; SCADA Command Injection (when combined with Dvvalin signal); Device Spoofing; and Hardware Fault (distinguished from cyber attacks by causal root-cause analysis using Granger causality testing on residual time series).

### 6.2.6 Module 6 — Response, Mitigation, and Visualisation

Module 6 (26) receives classified threat events from Module 5 and executes configurable automated response actions, including: generating structured alert records with MITRE ATT&CK for ICS technique mappings; streaming anomaly scores to a network-layer IDS (Suricata) via Lua script integration; logging all events to a Security Information and Event Management (SIEM) platform with tamper-evident hash-chain audit logging; generating automated incident reports conforming to CERT-In notification requirements; and triggering operator notifications via SMS and messaging applications.

## 6.3 Dvvalin — SCADA Network Intrusion Detection Engine (10)

The SCADA Network Intrusion Detection Engine (10) monitors industrial network traffic and detects anomalies in the cyber domain through a multi-layer detection approach:

(a) Rule-based detection: stateless rules flagging known attack patterns including unauthorised write commands to critical device registers, undocumented protocol function codes, and broadcast probe packets.

(b) Per-device behavioural profiling: statistical models of normal communication patterns for each device pair, including function code frequency distributions, register access ranges, and inter-arrival time distributions.

(c) Device timing fingerprinting: statistical fingerprinting of hardware-level response timing jitter per device. Authentication of device identity by Kolmogorov-Smirnov test comparison of observed response time distribution against the registered baseline distribution for the claimed device identifier.

(d) Graph-based lateral movement detection: representation of the substation communication network as a directed graph with devices as nodes and communication sessions as edges. Detection of new communication edges not present in the normal operational graph as indicators of lateral movement.

(e) Replay attack detection: payload hash caching, sequence number validation, and inter-arrival time uniformity testing for detection of scripted command replay attacks.

(f) Honeypot decoy devices: virtual industrial controller decoys integrated into the network. Any communication directed to a decoy device generates an immediate high-confidence alert with zero false positive rate.

(g) Man-in-the-middle detection: Cumulative Sum (CUSUM) change detection on per-session round-trip time measurements for detection of proxy insertion attacks adding measurable latency.

### 6.4 Valkyrie — Multi-Domain Fusion and Response Engine (40)

The Multi-Domain Fusion and Response Engine (40) combines anomaly scores from the active detection engines using confidence-weighted adaptive fusion:

*w_A = w_Aethelred * confidence_A (physics engine weight)*
*w_D = w_Dvvalin * confidence_D (network engine weight)*
*w_A, w_D = w_A,D / (w_A + w_D) (normalisation)*
*S_T = w_A * S_A + w_D * S_D (weighted fusion score)*
*If S_A > 0.5 AND S_D > 0.5: S_T = min(1.0, S_T * c_corr) (corroboration bonus)*

where w_Aethelred and w_Dvvalin are base weights; confidence_A and confidence_D are the confidence levels associated with each engine's current detection state; S_A and S_D are the respective anomaly scores; S_T is the final threat score; and c_corr is a configurable corroboration multiplier (default 1.25). The corroboration bonus reflects that simultaneous anomaly detection across both the physics and network domains provides substantially stronger evidence of a cyber-physical attack than either signal alone.

The Engine (40) maps each detected event to the MITRE ATT&CK for ICS framework and generates SHAP (SHapley Additive exPlanations) value decompositions for each alert, attributing the fractional contribution of each input feature to the final threat score. This explainability mechanism identifies the specific bus, measurement variable, and physical law violation responsible for triggering each alert.

### 6.5 Brynhildr — Thermodynamic Side-Channel Security Engine (30)

The Thermodynamic Side-Channel Security Engine (30) in a preferred embodiment monitors the thermal and electromagnetic emission signatures of grid field hardware. Physical modifications to device firmware, installation of hardware implants, or connection of hidden loads alter the power consumption profile and thermal emission patterns of affected devices in ways detectable through comparison against baseline thermodynamic profiles. In a preferred embodiment, Engine (30) employs infrared thermal imaging sensors co-located with distribution transformers and PLCs, comparing measured thermal signatures against expected thermal profiles derived from the known power consumption model of the device under monitored operating conditions.

### 6.6 Deployment Configurations

The system of the present invention supports three deployment configurations:

(a) Cloud-connected configuration: all engines deployed on a server at the substation or in a cloud environment; real-time monitoring with centralised SIEM aggregation.

(b) Edge-deployed configuration: the PINN model exported to an optimised inference format and deployed on an embedded computing device co-located with the substation; operates without network connectivity; synchronises events when connectivity is available.

(c) Federated configuration: multiple edge-deployed instances at different substations; model parameters aggregated using federated averaging, enabling collective learning from attack events observed at any substation without exchange of raw sensor data.

## 7. CLAIMS

**What is claimed is:**

### INDEPENDENT CLAIMS

**1.** A cyber-physical intrusion detection and response system for an electric power distribution network, comprising: a data ingestion module (21) configured to receive industrial network protocol traffic and extract physical state readings including voltage magnitude, voltage angle, active power, reactive power, and frequency at a plurality of network buses; a physics constraint engine (22) configured to solve AC power flow equations for the network topology and compute expected physical state values h(x_expected) for each bus; a Physics-Informed Neural Network (PINN) module (23) trained with a composite loss function comprising a data fidelity term L_data and a physics residual term L_physics encoding power flow governing equations, configured to predict physics-consistent expected sensor values; an anomaly detection module (24) configured to compute a False Data Injection residual r(z) as the absolute difference between observed sensor readings and expected values derived from said PINN module and said physics constraint engine, and to generate an anomaly flag when said residual exceeds a calibrated threshold; and a fusion and response engine (40) configured to combine anomaly scores from said anomaly detection module with anomaly scores from at least one additional detection domain into a unified threat score.

**2.** A method for detecting False Data Injection attacks on a power distribution network, comprising: receiving, by a processor, a plurality of observed sensor readings from network buses; computing, by a Physics-Informed Neural Network trained with a composite loss function L_total = lambda_d * L_data + lambda_p * L_physics + lambda_b * L_boundary, a physics-consistent expected state u_hat(x, t) for each bus; computing, by said processor, a physics oracle expected state h_oracle(x) by solving full nonlinear AC power flow equations for the network topology; computing a False Data Injection residual r(z) = |z_observed - h(x_expected)| wherein h(x_expected) is a weighted combination of u_hat and h_oracle; applying a statistical test to said residual to produce an anomaly score; and generating an alert when said anomaly score exceeds a Neyman-Pearson calibrated threshold.

**3.** A method for real-time electricity theft detection in a smart-metered power distribution network, comprising: receiving, by a processor, a feeder-level power injection measurement P_feeder_in from a substation metering point; receiving a plurality of individual customer meter readings P_meter_k; computing estimated technical line losses P_estimated_loss from network resistance parameters; computing a Non-Technical Loss ratio: NTL_ratio = (P_feeder_in - SUM_k(P_meter_k) - P_estimated_loss) / P_feeder_in; and generating a theft alert identifying the affected feeder segment when said NTL_ratio exceeds a configurable threshold, wherein said detection is independent of individual meter firmware and hardware integrity.

**4.** A multi-domain anomaly fusion method for cyber-physical intrusion detection, comprising: receiving a physics-domain anomaly score S_A from a Physics-Informed Neural Network anomaly detection engine; receiving a network-domain anomaly score S_D from a SCADA network intrusion detection engine; computing respective confidence-adjusted weights w_A = w_base_A * conf_A and w_D = w_base_D * conf_D; normalising said weights; computing a weighted fusion score S_T = w_A * S_A + w_D * S_D; applying a corroboration bonus to said fusion score when both S_A and S_D

independently exceed a corroboration threshold; and classifying said fusion score into threat levels and triggering automated response actions corresponding to the classified threat level.

## DEPENDENT CLAIMS

**5.** The system of claim 1, wherein said PINN module implements a Graph Attention Network (GAT) representing the power network as a graph with buses as nodes and transmission lines as edges, wherein the physics residual loss is computed per node and aggregated across the graph.

**6.** The system of claim 1, wherein said PINN module implements Bayesian uncertainty quantification using Monte Carlo Dropout, performing a plurality of stochastic forward passes at inference time to produce a mean prediction and a per-output standard deviation, and wherein the anomaly detection module triggers an alert only when the observed reading deviates from the predicted mean by more than a configurable number of standard deviations.

**7.** The system of claim 1, further comprising a multi-timescale detection module implementing a first detection window of a configurable small number of consecutive readings for detecting transient FDI attacks, and a second detection window of a configurable larger number of readings for detecting gradual electricity theft and slow-ramp manipulation attacks, wherein the anomaly score is the maximum of scores computed over both windows.

**8.** The system of claim 1, further comprising a SCADA network intrusion detection engine configured to monitor industrial protocol traffic including at least one of Modbus TCP, DNP3, IEC-61850 GOOSE, IEC-61850 MMS, and OPC-UA; performing per-device behavioural profiling; and detecting at least one of unauthorised command injection, device impersonation, replay attacks, and lateral movement.

**9.** The system of claim 8, wherein said SCADA network intrusion detection engine performs device identity authentication by statistical fingerprinting of hardware-level response timing jitter, comparing observed response time distributions against registered baseline distributions using a Kolmogorov-Smirnov test.

**10.** The system of claim 8, wherein said SCADA network intrusion detection engine maintains a directed graph of normal communication patterns between devices, and generates an immediate alert upon detection of a new communication edge absent from said normal communication graph.

**11.** The system of claim 8, wherein said SCADA network intrusion detection engine includes one or more virtual decoy industrial controller devices integrated into the network, wherein any communication directed to a decoy device generates a high-confidence alert.

**12.** The system of claim 1, further comprising a Man-in-the-Middle detection module implementing a Cumulative Sum (CUSUM) change detection algorithm on per-session round-trip time measurements, generating an alert when the cumulative sum statistic exceeds a configurable decision boundary.

**13.** The system of claim 1, further comprising a thermodynamic side-channel security engine configured to monitor thermal and electromagnetic emission signatures of grid field hardware and detect deviations from baseline thermodynamic profiles indicative of firmware modification, hardware implant installation, or connection of undeclared loads.

**14.** The method of claim 2, further comprising computing voltage stability L-indices at load buses as $L_j = |1 + SUM_i(F_{ji} * V_i / V_j)|$ where $F = -[Y_{LL}]^{(-1)} * [Y_{LG}]$, and generating operational stability alerts when said L-indices approach unity.

**15.** The method of claim 2, further comprising applying SNR-based feature selection to detection input features, wherein features are ranked by the ratio of between-class to within-class variance, and low-ranking features are excluded from the anomaly detection computation.

**16.** The method of claim 2, wherein the PINN module is pre-trained on IEEE standard power network test cases and fine-tuned on target network operational data using transfer learning with selective layer freezing, wherein layers encoding universal power flow physics are frozen and layers encoding network-specific topology patterns are fine-tuned.

**17.** The fusion method of claim 4, further comprising generating an explanatory decomposition of the threat score using SHAP (SHapley Additive exPlanations) values attributing the fractional contribution of each input feature to the computed threat score.

**18.** The fusion method of claim 4, further comprising automatic mapping of each classified threat event to a corresponding tactic and technique in the MITRE ATT&CK for ICS framework, and automatic generation of a structured incident report conforming to CERT-In mandatory notification format.

**19.** The system of claim 1, wherein said PINN module is deployed in an edge-computing configuration on an embedded device co-located with the substation, operating without continuous network connectivity, with the PINN model exported to an optimised inference format and executed by a hardware-accelerated inference engine.

**20.** The system of claim 1, wherein a plurality of instances of the system are deployed at respective substations of a distribution utility, and model parameters are periodically aggregated using a federated learning protocol that combines locally computed model updates without exchange of raw sensor data, preserving

data confidentiality of each substation while enabling collective improvement of detection capability across the distribution network.

## 8. ABSTRACT OF THE DISCLOSURE

A multi-domain cyber-physical intrusion detection and response system for electric power distribution networks is disclosed. The system comprises: a Physics-Informed Neural Network (PINN) engine that validates observed sensor readings against expected values derived from AC power flow equations, detecting False Data Injection attacks and electricity theft that are undetectable by network-layer monitoring systems; a SCADA network intrusion detection engine covering Modbus TCP, DNP3, IEC-61850, and OPC-UA protocols with per-device behavioural profiling and device timing fingerprinting; and a multi-domain fusion engine combining physics-domain and network-domain anomaly scores with confidence-weighted adaptive fusion and corroboration bonus. The PINN is trained with a composite loss L_total = lambda_d * L_data + lambda_p * L_physics + lambda_b * L_boundary embedding Kirchhoff's Current Law and AC power flow constraints. FDI residuals are computed as r(z) = |z_observed - h(x_expected)| using a dual oracle combining the PINN prediction and a PyPSA physics oracle. Electricity theft is detected via a Non-Technical Loss ratio computed from feeder-level and meter-level power measurements. The system generates SHAP-explainable alerts with MITRE ATT&CK for ICS mappings and automated CERT-In incident reports, and supports cloud-connected, edge-deployed, and federated deployment configurations.

*Keywords: cyber-physical security; power grid; intrusion detection; Physics-Informed Neural Networks; False Data Injection; electricity theft; SCADA; anomaly detection; multi-domain fusion.*

---

## DECLARATION

I/We hereby declare that the invention as disclosed in the Complete Specification filed herewith has not been claimed in any other specification filed in India. I/We declare that I am/we are the true and first inventor(s) of the invention described herein.

Signature of Applicant/Inventor: _____

Name: [FULL LEGAL NAME]

Date: _____

Place: _____

---

*Filed at the Indian Patent Office, _____ Office*

*Application No.: ____/____/2026 | Date of Filing: _____*